Deep Reinforcement Learning for IRS-Based Secure Wireless Communications Under Imperfect Eavesdropper CSI

CY315 — Wireless and Mobile Security | Project Proposal | Track 2: Implementation & Optimization

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***Abstract*—Physical Layer Security (PLS) exploits the physical properties of wireless channels to achieve information-theoretic secrecy without relying on computational hardness assumptions. Intelligent Reflecting Surfaces (IRS) have emerged as a powerful tool for PLS by reshaping the propagation environment to favour legitimate receivers over eavesdroppers. Existing Deep Reinforcement Learning (DRL) approaches to IRS-aided PLS, including the prominent work of Yang et al. [1], assume that the transmitter has access to the eavesdropper's channel state information (CSI). This assumption is unrealistic for passive eavesdroppers who never participate in channel estimation. This project implements the Yang et al. baseline system and extends it by removing the eavesdropper CSI requirement entirely. We replace perfect Eve CSI with a worst-case secrecy model under bounded channel uncertainty and retrain the DRL agent under this more realistic setting. Comparative evaluation across varying SNR levels and IRS element counts will demonstrate that the proposed system maintains robust secrecy performance where the baseline degrades.**

***Index Terms*—Physical layer security, intelligent reflecting surface, deep reinforcement learning, DDPG, eavesdropper CSI, secrecy rate, robust beamforming.**

I. Problem Statement

When two parties communicate wirelessly, the transmitted signal propagates in every direction and can be captured by any passive listener within range. This eavesdropping threat exists at the physical layer, before any application-level encryption takes effect. Physical Layer Security (PLS) addresses this threat by exploiting the structure of the wireless channel itself rather than relying on computational complexity.

The core principle of PLS is that if the legitimate receiver (Bob) has a stronger channel than the eavesdropper (Eve), it is theoretically possible to transmit information at a rate that Bob can decode but Eve cannot [2]. The achievable secrecy rate quantifies this advantage:

$$R_s = \max\{\log_2(1+\mathrm{SNR}_{\mathrm{Bob}}) - \log_2(1+\mathrm{SNR}_{\mathrm{Eve}}),\ 0\} \tag{1}$$

A positive R_s guarantees the existence of a coding scheme under which Eve cannot recover the transmitted message regardless of her computational resources. The engineering challenge is constructing a wireless environment in which this condition holds reliably, especially when Eve's location and channel are unknown.

Intelligent Reflecting Surfaces (IRS) are a promising tool for this purpose. An IRS is a planar array of passive reflecting elements, each capable of independently shifting the phase of the impinging signal. The IRS phase shift matrix is defined as:

$$\boldsymbol{\Theta} = \mathrm{diag}\big(e^{j\theta_1},\, e^{j\theta_2},\, \ldots,\, e^{j\theta_N}\big) \tag{2}$$

By optimising $\boldsymbol{\Theta}$, the IRS steers reflected energy constructively toward Bob while creating destructive interference at Eve, widening the SNR gap without consuming additional transmit power. The challenge is determining the optimal phase configuration in real time as channels fluctuate.

Classical convex optimisation methods operate on static channel snapshots and incur prohibitive computational overhead under dynamic conditions. Deep Reinforcement Learning (DRL) addresses this by training an agent to learn a policy that maps observed channel states to near-optimal phase configurations through interaction with the environment — no iterative solving is required at runtime.

Yang et al. [1] proposed a DRL-based solution to this problem using a Deep Deterministic Policy Gradient (DDPG) agent and demonstrated strong performance over conventional optimisation baselines. However, their formulation includes a critical unrealistic assumption: the transmitter (Alice) is given Eve's instantaneous CSI as part of the state vector. In practice, passive eavesdroppers do not transmit pilot signals and therefore their channels cannot be estimated at Alice. This is the gap our project addresses.

II. Proposed Contribution

We implement the Yang et al. [1] system as our baseline, adopting their system model, channel configuration, and Markov Decision Process (MDP) formulation. Our objective is not exact architectural replication but to establish a fair reference point before applying our contribution.

Our contribution removes Eve's CSI from the system entirely. In the modified system, the DRL agent's state vector contains only the Alice–Bob direct channel and the Alice–IRS–Bob cascaded channel. The reward function is redesigned to compute a worst-case secrecy rate by sampling K random Eve channel realisations within a statistical uncertainty region and rewarding the minimum secrecy rate across all samples:

$$r_{\mathrm{wc}} = \min_{k \in \{1,\ldots,K\}} \max\{\log_2(1+\mathrm{SNR}_{\mathrm{Bob}}) - \log_2(1+\mathrm{SNR}_{\mathrm{Eve},k}), 0\} \tag{3}$$

where each $\mathrm{SNR}_{\mathrm{Eve},k}$ corresponds to a sampled Eve channel drawn from the same Rayleigh fading distribution. The agent is trained to find IRS phase configurations that are robust across this uncertainty set, rather than optimal only for a single known Eve position.

This produces a security guarantee that is honest: the system performs well not because we assumed a particular eavesdropper condition, but because the agent has learned robustness against adversarial channel uncertainty.

III. System Model

The simulation environment consists of four entities as shown in Fig. 1.

- **Alice**: A base station with $N_t = 4$ transmit antennas.
- **Bob**: A single-antenna legitimate receiver.
- **Eve**: A single-antenna passive eavesdropper whose channel is unknown to Alice.
- **IRS**: A uniform linear array of $N \in \{32, 64\}$ passive reflecting elements.

All channels follow Rayleigh fading, i.e., channel coefficients are drawn from $\mathcal{CN}(0,1)$ and are refreshed every episode to evaluate agent adaptability. The received SNR at receiver $x \in \{\mathrm{Bob},\ \mathrm{Eve}\}$ is:

$$\mathrm{SNR}_x = \frac{P_t \left|\mathbf{h}_{Ax}^H \mathbf{w} + \mathbf{h}_{Ix}^H \boldsymbol{\Theta}\, \mathbf{H}_{AI}\, \mathbf{w}\right|^2}{\sigma^2} \tag{4}$$

where $\mathbf{w} \in \mathbb{C}^{N_t}$ is Alice's transmit beamforming vector, $\mathbf{H}_{AI} \in \mathbb{C}^{N \times N_t}$ is the Alice–IRS channel matrix, $\mathbf{h}_{Ix}$ is the IRS–receiver channel vector, $\mathbf{h}_{Ax}$ is the direct Alice–receiver channel, and σ^2 is the noise power.

Fig. 1. System model. Alice transmits to Bob via direct and IRS-reflected paths. Eve captures both paths passively. The DRL agent controls the IRS phase shifts $\boldsymbol{\Theta}$ to maximise R_s.

The DRL agent's action at each step is the tuple $(\boldsymbol{\theta}, \mathbf{w})$ comprising the N IRS phase angles and the beamforming vector. In the baseline, the state vector includes Eve's CSI. In our modified system Eve's channel vectors are excluded from the state, reducing the state dimension from $2N_t + 2NN_t + 2N + 2N_t + 2N$ to $2N_t + 2NN_t + 2N$.

IV. Baseline Reference

Our baseline is the system proposed by Yang et al. [1], which trains a DDPG agent with a continuous action space covering IRS phase shifts and transmit beamforming. We adopt their system model, Rayleigh fading channel setup, and MDP formulation. Our focus is on the contribution that goes beyond this baseline — eliminating the Eve CSI requirement — rather than replicating every architectural detail of their agent implementation.

V. Methodology

The project proceeds in three sequential phases.

Phase 1 — Baseline Implementation. We implement the Yang et al. system model in Python using PyTorch. A DDPG agent is trained with Eve's CSI included in the state vector and the exact secrecy rate (1) as the reward signal. Secrecy rate vs. transmit SNR curves for $N = 32$ and $N = 64$ IRS elements are generated as the reference baseline for comparison.

Phase 2 — Contribution: Removing Eve CSI. The environment is modified in two ways. First, Eve's channel vectors $\mathbf{h}_{AE}$ and $\mathbf{h}_{IE}$ are removed from the agent's state vector. Second, the reward function is replaced with the worst-case secrecy formulation of (3), sampling $K = 50$ random Eve channel realisations per training step. The agent is retrained from scratch under this new specification.

Phase 3 — Comparative Evaluation. Both systems are evaluated under identical channel conditions. The following comparative results are generated:

1) Secrecy rate vs. transmit SNR: baseline vs. proposed system.
2) Secrecy rate vs. number of IRS elements ($N = 16, 32, 64$).
3) Training convergence curves: reward per episode.
4) Secrecy rate vs. Eve channel uncertainty region size.

VI. Tools and Environment

TABLE I
Development Environment

Component	Tool
Programming language	Python 3.10+
DRL framework	PyTorch
Numerical computation	NumPy, SciPy
Visualisation	Matplotlib
Report	(Overleaf, IEEE format)
Version control	Git

VII. Expected Outcomes

The baseline system will achieve a higher peak secrecy rate due to the informational advantage of perfect Eve CSI. When evaluated under realistic conditions (Eve CSI withheld), the baseline policy degrades significantly because it was never trained to be robust to CSI uncertainty.

Our proposed system accepts a moderate reduction in peak secrecy rate as the cost of robustness. Under realistic evaluation conditions, however, our system will maintain substantially higher and more consistent secrecy performance than the baseline. The central claim is:

The baseline achieves higher peak performance under an assumption that cannot be satisfied in practice. Our system achieves reliable security under an assumption that is always satisfied in practice.

VIII. Conclusion

This project implements and extends a state-of-the-art DRL-based IRS security system by removing a critical and unrealistic assumption — knowledge of the eavesdropper's channel state information. The proposed worst-case reward formulation enables the DDPG agent to learn IRS phase configurations that maintain positive secrecy rate even when the eavesdropper's location and channel are entirely unknown to the transmitter. This represents a concrete step toward physically deployable IRS-aided physical layer security systems suitable for real wireless threat environments.

References

[1] H. Yang, Z. Xiong, J. Zhao, D. Niyato, L. Xiao, and Q. Wu, "Deep Reinforcement Learning-Based Intelligent Reflecting Surface for Secure Wireless Communications," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 375–388, Jan. 2021.

[2] M. Cui, G. Zhang, and R. Zhang, "Secure Wireless Communication via Intelligent Reflecting Surface," *IEEE Wireless Commun. Lett.*, vol. 8, no. 5, pp. 1410–1414, Oct. 2019.

[3] Y. Zhang *et al.*, "Deep Reinforcement Learning for Secrecy Energy Efficiency Maximization in RIS-Assisted Networks," *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 12413–12418, 2023.

[4] "RIS-Assisted Physical Layer Security: A Comprehensive Survey," *IEEE Access*, 2024.